

Problem 6

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Problem 1 Let

$$h(u) = \begin{cases} \frac{u^2 - 5u + 4}{u - 4} & \text{if } u < 4 \\ \frac{-\sqrt{u+4}}{u-6} & \text{if } u \geq 4, u \neq 6. \end{cases}$$

(a) What is the domain of h ?

Explanation. For values of $u < 4$ we check $\frac{u^2 - 5u + 4}{u - 4}$. This piece of the function is defined everywhere on $(-\infty, 4)$

For values of $u \geq 4$ we check $\frac{-\sqrt{u+4}}{u-6}$. This piece of the function is defined everywhere except $u = 6$. So h is defined on $[4, 6) \cup (6, \infty)$. The domain is $(-\infty, 6) \cup (6, \infty)$

(b) Find all vertical asymptotes of h . **EXPLAIN.**

Explanation. The only candidates are $u = 6$ and $u = 4$.

$\lim_{u \rightarrow 6^+} \frac{-\sqrt{u+4}}{u-6} = -\infty$ because it is of the form $\frac{\#}{0}$, the numerator negative, the denominator positive and

$$\lim_{u \rightarrow 4^-} \frac{u^2 - 5u + 4}{u - 4} = \lim_{u \rightarrow 4^-} \frac{(u-1)(u-4)}{u-4} = \lim_{u \rightarrow 4^-} (u-1) = 3$$

$$\lim_{u \rightarrow 4^+} \frac{-\sqrt{u+4}}{u-6} = \frac{-\sqrt{8}}{-2} = \sqrt{2}$$

The only vertical asymptote is $u = 6$, because $\lim_{u \rightarrow 6^+} h(u) = -\infty$.

(c) Find all horizontal asymptotes of h . **EXPLAIN.**

Explanation.

$$\begin{aligned}\lim_{u \rightarrow \infty} \frac{-\sqrt{u+4}}{u-6} &= \lim_{u \rightarrow \infty} \frac{-\frac{\sqrt{u+4}}{\sqrt{u}}}{\frac{u-6}{\sqrt{u}}} \\ &= \lim_{u \rightarrow \infty} \frac{-\sqrt{\frac{u+4}{u}}}{\frac{u}{\sqrt{u}} - \frac{6}{\sqrt{u}}} \\ &= \lim_{u \rightarrow \infty} \frac{\sqrt{1 + \frac{4}{u}}}{\sqrt{u} - \frac{6}{\sqrt{u}}} \\ &= 0\end{aligned}$$

because it is of the form: $\frac{\#}{\infty}$

This indicates that $y = 0$ is a horizontal asymptote.

Checking as u approaches $-\infty$ we have:

$$\lim_{u \rightarrow -\infty} \frac{u^2 - 5u + 4}{u - 4} = \lim_{u \rightarrow -\infty} \frac{(u-1)(u-4)}{u-4} = \lim_{u \rightarrow -\infty} (u-1) = -\infty$$

Therefore, there is no horizontal asymptote as u approaches $-\infty$

There is one horizontal asymptote at $y = 0$, because $\lim_{u \rightarrow \infty} h(u) = 0$.

(d) List the **Intervals of Continuity** for the function h .

Explanation. Remember that finding Intervals of Continuity means to find the largest such intervals. For values of $u < 4$ we check $\frac{u^2 - 5u + 4}{u - 4}$. This is a piece of a rational function and is continuous everywhere on $(-\infty, 4)$.

For values of $u > 4$ we check $\frac{-\sqrt{u+4}}{u-6}$. This piece of the function is continuous everywhere where defined: $(4, 6), (6, \infty)$

We need to check $u = 4$ for continuity. A function is continuous at a point a if:

$$\begin{aligned}f &\text{ is defined at } x = a \\ \lim_{x \rightarrow a} f(x) &\text{ exists} \\ \lim_{x \rightarrow a} f(x) &= f(a)\end{aligned}$$

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h is defined at $u = 4$, $h(4) = \frac{-\sqrt{4+4}}{4-6} = \sqrt{2}$

Next we check if $\lim_{u \rightarrow 4} h(u)$ exists. That is, does $\lim_{u \rightarrow 4^-} h(u) = \lim_{u \rightarrow 4^+} h(u)$?

$$\lim_{u \rightarrow 4^-} h(u) = \lim_{u \rightarrow 4^-} \frac{u^2 - 5u + 4}{u - 4}$$

This is of the form: $\frac{0}{0^-}$ so we need to do more algebra

$$\begin{aligned} &= \lim_{u \rightarrow 4^-} \frac{(u-4)(u-1)}{(u-4)} \\ &= \lim_{u \rightarrow 4^-} (u-1) \\ &= 4-1 \\ &= 3 \end{aligned}$$

$$\begin{aligned} \lim_{u \rightarrow 4^+} h(u) &= \lim_{u \rightarrow 4^+} \frac{-\sqrt{u+4}}{u-6} \\ &= \frac{-\sqrt{4+4}}{4-6} \\ &= \frac{-\sqrt{8}}{-2} \\ &= \sqrt{2} \end{aligned}$$

$$\lim_{u \rightarrow 4^-} h(u) \neq \lim_{u \rightarrow 4^+} h(u) \implies \lim_{u \rightarrow 4} h(u) \text{ does not exist}$$

Thus $h(u)$ is not continuous at $u = 4$. Remark: Since $\lim_{u \rightarrow 4^+} h(u) = h(4)$, h is right continuous at $u = 4$. Its intervals of continuity are $(-\infty, 4)$, $[4, 6)$, $(6, \infty)$